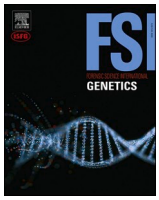
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The advent of forensic DNA databases: It's time to agree on some international governance principles!

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ABSTRACT

National forensic DNA databases are a valuable investigative tool, that have the potential to increase the efficacy of criminal investigations. Their unfettered expansion in recent years raises unsettling ethical issues that require close attention. DNA database expansion threatens the rights to privacy, non-discrimination, and equality, and can undermine public trust in government. This perspective piece relies on data from an international mapping study of Forensic DNA Databases to document the expansion of these databases, highlight the ethical issues they raise, and propose key recommendations for more responsible use of this infrastructure.

1. Background

DNA technology has revolutionized the science of forensics. From the introduction of 'genetic fingerprinting' in high profile criminal cases in the mid 1980s, to the now routine use of 'genetic profiling,' DNA identification has become an integral part of criminal investigations worldwide. By comparing DNA evidence found at crimes scenes with the DNA profile of individuals stored in DNA databases, investigators are now better able to identify potential suspects and solve crimes. In large part, the establishment of national forensic DNA databases (DNA databases), that contain the profiles of suspects and individuals, has facilitated this revolution. As with many advancements in law enforcement technologies, however, the unfettered expansion of DNA databases in recent years has raised complex legal and ethical questions. This paper proceeds by documenting the advent of forensic DNA databases, highlighting the ethical, legal, and social implications of their expansion, and recommends policy frameworks for more responsible use of DNA database technology.

2. Advent of forensic DNA databases

The inception of forensic genetics in the 1980s, heralded the development of government DNA databases. In 1995, the United Kingdom established the National DNA Database (NDAD)—the world's first Combined DNA Index System (CODIS)—based DNA database. Since then, the number of countries operating DNA databases and the number of profiles contained within them has grown exponentially [1]. Most countries around the world have now established state DNA databases [2].

As reliance on DNA databases has grown, investigators have steadily expanded the number of DNA profiles stored in national databases. At present, China, followed by the United States and the United Kingdom operate the largest forensic DNA databases [3]. As of October 2021, the U.S. National DNA Index (NDIS) contained over 14,836,490 offender profiles, 4513,955 arrestee/detainee profiles and 1144,255 other forensic profiles [4]. Our forensic policy map reveals that in 9 out of 84 countries operating forensic databases, authorities have now accumulated DNA profiles for more than 5 % of their populations (see Fig. 1) [5].

The scope of DNA databases has expanded as governments have gathered the genetic information of more individuals and have relied on technologies that produce increasingly detailed genetic profiles. For the most part, these databases have relied on single tandem repeats (STRs), a sequence of two or more DNA bases that is repeated numerous times on a chromosome. While Government DNA databases typically use 15–20 STRs to create a profile for comparison with crime scene profiles, there has been a trend amongst governments to increase the number of STR being used [2]. The greater the number of STR loci used for typing, the greater the discrimination value as the likelihood that a single individual will have an identical STR profile (the exact same number of repeat units for all the STRs being analyzed) with another individual taken at random in the population becomes extremely rare. Countries have also gone beyond the core CODIS based autosomal loci, to include STRs on the Y chromosome (Y-STRs), and single nucleotide polymorphisms (SNPs). A SNP is a genomic variant at a single base position in the DNA and is the most common type of genetic variation among people. When

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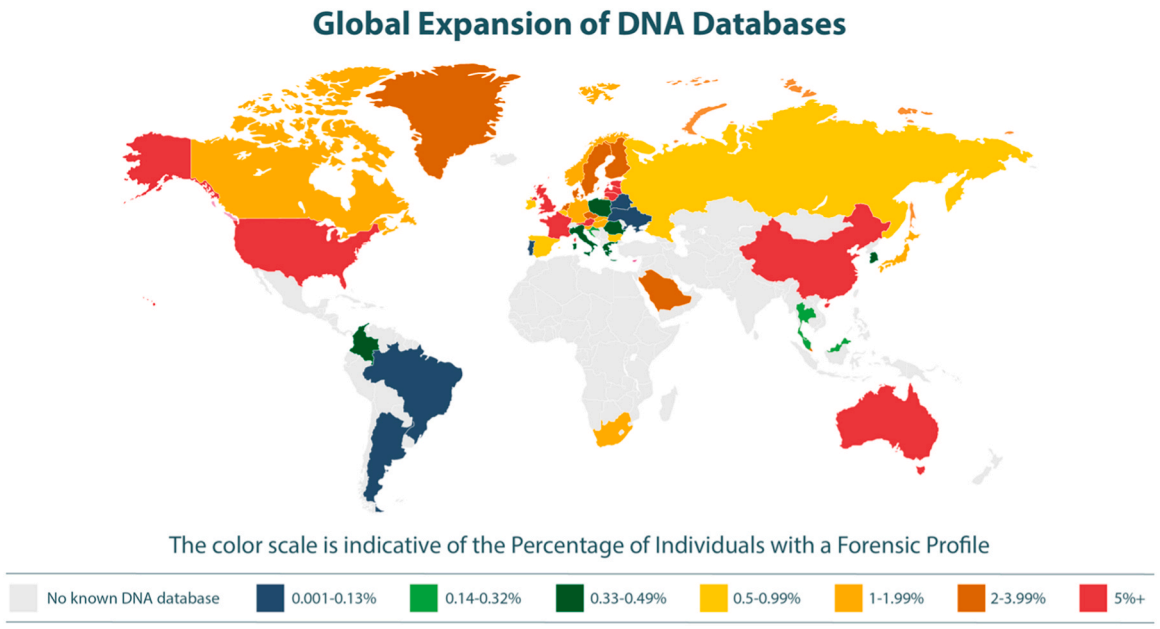


Fig. 1. Expansion of DNA databases.

used in investigations, SNP scans reveal significant genetic information not available through STRs in government databases, allowing law enforcement to identify potential suspects with greater accuracy [6,7].

3. Function creep

Although DNA databases are vital tools for solving crimes, innovations to the underlying technology, and novel uses of DNA data raise ethical concerns that demand careful attention. Governmental DNA databases were originally strictly regulated and used to solve serious crimes. These databases gathered limited STR profiles, restricted to non-coding regions, and included limited health information. But, over time, law enforcement has begun employing mechanisms to expand their DNA marker profile pool, to investigate lesser transgressions, and profile groups beyond convicted felons [8]. When technologies such as DNA databases are used for purposes beyond what was originally intended, it is known as function creep. Function Creep risks not only jeopardizing the right to privacy and other human rights, but can erode trust in government.

While the initial use of DNA in forensics investigation was considered quite controversial, there has been a trend amongst countries to relax legal restrictions on the content and use of DNA databases [8]. DNA databases are now used to profile not just convicted felons, but arrestees/detainees and innocent people as well. Worldwide, police services have also turned to commercial or public research databases to expand their profile pool. Studies show that additional databases, such as the Y-chromosome haplotype database have been developed and used to target ethnic groups [9]. While, techniques such as Y-STRs and SNPs, may expand the size of a pool, they also provide investigators with more personal information about a suspect than traditional STRs.

Law enforcement has also started submitting crime scene samples to commercial ancestry testing databases when comparisons within government databases yield no leads [10]. Commercial DNA databases are sometimes able to identify genetic relatives via a partial match. Once a partial match is obtained, investigators, will employ techniques such as Familial Searching, and forensic genetic genealogy (FGG) to identify potential suspects. They then surreptitiously collect DNA from a suspect's family members, develop STR profiles and compare them with the crime scene CODIS profile. Once a match is found, law enforcement is able to obtain a warrant and legally collect a DNA sample [6]. DNA, however, is a "trove of information," and if the expansion of use to non-governmental databases, is not strictly regulated, DNA can be misused in a manner highly detrimental to the fundamental rights of suspects, their families and broader population groups.

While DNA databases can increase the efficacy of criminal investigations, it is still necessary that countries safeguard against the human rights implications they pose. Some countries, such as Germany, have restricted the expansion to private and public databases, but others, such as Canada and the United States, have implemented legislation to expand the forensic use of DNA beyond government DNA databases [10]. The following section considers some of the legal and ethical concerns of DNA database expansion.

3.1. *Misuse & the right to privacy*

Without regulatory safeguards, DNA databases can be misused in ways that undermine fundamental rights of people. DNA profiling kits utilizing SNP arrays are designed to provide phenotype and kinship information [11]. As explained above, the adoption of additional markers (SNPs, Y-STRs, and mtDNA) and new techniques such as whole genome sequencing (WGS) may expand the breadth of DNA databases. But the expansion of the types of DNA profiles in forensic databases also provides law enforcement with more sensitive information about a person, including their ethnicity, gene expression levels, and potential medical information [12,13]. The threat these databases pose to privacy are thus considerable, since data can be used by law enforcement for purposes beyond forensic investigations and may be collected without the consent of a suspect.

Studies show that investigators are increasingly utilizing genetic information, in a process called forensic genetic genealogy (FGG), from direct-to-consumer companies to identify suspects in criminal cases. In comparison to CODIS's 15–20 STRs, consumer ancestry databases include over 600,000–1million SNPs, which can reveal significant information about a person's health [1]. Understood, as a SNP-based relative matching technique, where a profile is combined with family tree research, FGG, can be a particularly fruitful tool for developing leads in criminal investigations. As of September 2021, the use of FGG has led to the discovery of over 150 suspects of murder and sexual assault in the United States [6].

At the same time however, FGG risks involuntarily involving the

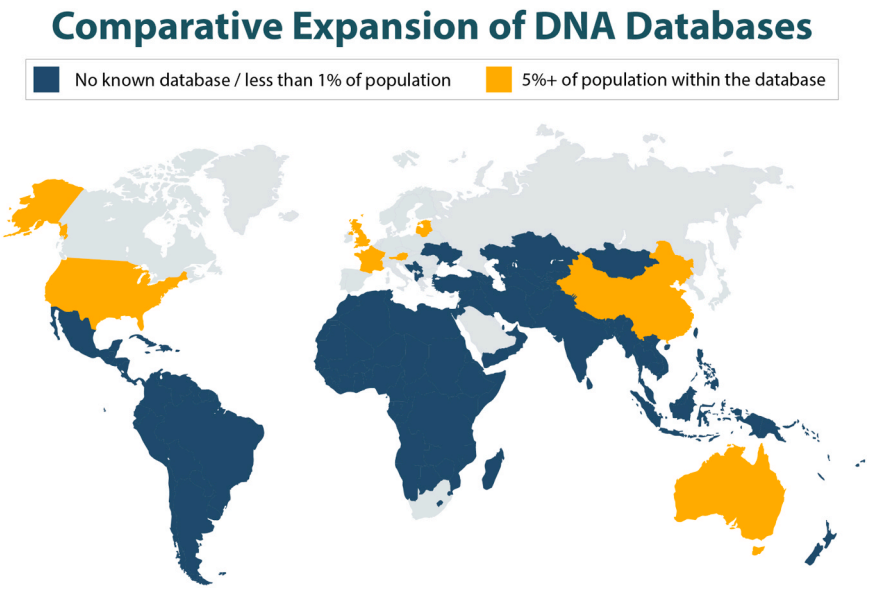


Fig. 2. comparative expansion of DNA databases.

family members of suspects in investigations—potentially violating the fundamental rights of suspects, and of their family members. Some scholars claim that the impact of these technologies is so grave for genetic relatives, that their collective right to privacy, freedom from arbitrary search and presumption of innocence are violated by their use. Even as courts, including the European Court of Human Rights, have held that the use of private DNA databases to identify genetic relationships between individuals violates the right to privacy [14], the use of FGG is increasing in Europe and North America.

Privacy concerns are also compounded by the risk of data breach. Recently, information from more than 92 million accounts from the genealogy and DNA testing service My Heritage were found on a private server [15]. Although DNA data itself was not breached, the incident highlighted the potential of data breaches to reveal a person's sensitive information to the world. Private databases lack stringent security measures to guard against possible data breaches. While government databases are subject to security regulations, inherent vulnerabilities in software tools could render them susceptible to cyber-attacks as well. The possibility of such abuses could undermine trust in the direct-to-consumer genetic industry and in government as well.

3.2. *Racial profiling*

The expansion of DNA databases may also exacerbate discrimination and growing racial and ethnic disparities. Studies show that current technologies, including CODIS profiles are unreliable for racial classifications [12,16]. Yet, evidence from the US suggests that forensic DNA databases are growing to mirror disparities in arrest practices, which have already been established to be biased [16]. People of color make up almost half of the national DNA database (47 %), even though these groups make up only a little over a third (39 %) of the country's population [17].

There is variability amongst databases worldwide, and some include the profiles of arrestees, suspects, and even innocent people. As technology advances, we will need to consider the impact that the use of databases in conjunction with future technologies (facial recognition, rapid DNA kits, and artificial intelligence), may have on groups already marginalized by the criminal system. Function creep is already a concern in some jurisdictions, where police services are using advanced technologies, such as, whole genome sequencing to analyze degraded crime scene DNA samples. The depth of information obtained from whole genome sequencing has dwarfed SNP testing, allowing investigators to access biogeographical, phenotypic and other health and demographic data [18]. A growing number of studies show that advanced technologies, including predictive policing tools perpetuate systemic racism, and yet we still know very little about how they work, who is using them, and for what purpose [19]. As technology continues to evolve, legislators will have to seriously consider the implications of DNA databases on marginalized populations, to find ways to make the system more humane and equitable for these groups.

3.3. *Global divide*

The rapid expansion of DNA databases in the global north also raises concerns about discrepancies in access to scientific information among countries. Our study shows that the United States and China have established the largest databases, collectively holding around one hundred million DNA profiles. In contrast, few countries in the global south have even started to establish forensic DNA databases (Fig. 2). DNA technology is expensive, and high costs could hinder the spread of these innovations to less wealthy nations. This disparity may have important political and social consequences. For one, high income countries could use these technologies to keep individuals from lower income nations from entering their borders. Countries with advanced DNA databases may require immigrants and visitors to provide genetic samples upon entry. Europe, for example, has already established a two-speed system for travelers depending on whether their countries provide biometric passports or not. This type of data gathering could widen the gap between wealthier countries that maintain sophisticated DNA databases and those that have not yet developed similar infrastructures. It could also propel nations to adopt similar measures.

In the name of surveillance and consolidation of power, countries in the global south could be impelled to outdo the global north, but at a cost to their own citizens' rights. In 2015, for example, Kuwait became the first country to require all citizens and visitors to submit their DNA samples. [20] While this law was arguably introduced to allow identification of potential terrorists, it shows how any law mandating universal data collection can be abused [20]. Following condemnation from UNESCO, the government revoked the law.

4. Moving towards more responsible use of forensic DNA databases

National forensic DNA databases are a valuable investigative

Table 1
Points to consider.

Points to Consider
1. Ensure that there are standardized, compulsory quality assurance programs for forensic laboratories and accreditation programs for laboratory technicians.
2. Provide mandatory training to all law enforcement personnel likely to be involved in the collection, storage, and use of DNA samples and data in order to ensure traceability, data integrity safe-guards, ethical aspects of forensic DNA and permissible use.
3. DNA samples collected from individuals neither suspected, nor convicted, of a criminal act should not be used for the purpose of identifying a suspect. In the event DNA samples are used for identifying human remains, informed consent of the sample provider should first be obtained.
4. Undertake privacy risk assessments of forensic DNA databases and associated genetic laboratories on a regular basis, and in the aftermath of data breaches. In the event of a breach, all individuals whose profile is held in a database should be notified within a reasonable time-period.
5. Ensure the transparency of these databases, through a regularly updated public website, disclosing lay information, including, the number of profiles and their demographics, scope of use, and any breaches.
6. Prohibit the ad-hoc use of new investigative techniques using DNA databases without prior transparent assessment of the process, public consultation, and the development of a proper regulatory framework.
7. Conduct further studies on the impact of DNA database expansion (both in terms of content and scope of use) on regional, national, and international equity.

resource that have the potential to improve the efficacy of criminal investigations and clear the names of wrongly accused individuals. However, the unfettered enthusiasm of law enforcement authorities and policymakers for rapidly expanding DNA databases, without addressing the complex ethical challenges they raise, is unsettling. Stakeholders need to assume greater responsibility for the governance of forensic databases, or else tools intended to improve the capacity to solve crime could become a source of oppression for the people they serve. While Interpol promulgated Best Practices for DNA Databases in 2019 [21] the recommendations are mostly directed at improving the quality and efficiency of the process. Topics like ethics, governance and human rights are only mentioned as relevant considerations, but Interpol did not recommend any minimum safeguards.

We propose that policymakers and custodians of databases worldwide consider the following broadly formulated priorities as a starting point for a more optimal governance framework (see list below):

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Nicole Palmour: Conceptualization, Writing – review & editing. **Diya Uberoi:** Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing. **Yann Joly:** Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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